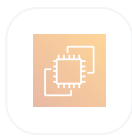


LionX compute: Amazon ECS

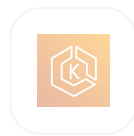
ADR-0001. The compute platform decision behind the AWS migration.



ECS Fargate



ECS on EC2



EKS

proposed

2026-06-26 Sofiane DJERBI, Bayarbileg BATBILEG, Rui SOUSA CALDAS, Laurent QUILLÉROU

The decision, and what drives it

One choice to make: where LionX compute runs on AWS.

THE WORKLOAD



8 services

long-running, plus cron



8 vCPU, 32 GB

per environment



5 environments

integration to production

Networking, CI/CD, observability and IAM all hang off this choice.

WHAT DRIVES THE DECISION

1

End-2026 cutover.

The platform has to fit the migration runway, not the other way around.

2

A rewrite either way.

The deployment layer is rewritten regardless. The only question is the target.

3

Audit surface stays minimal.

Every extra component is more to audit, scan, log and upgrade.

4

A single workload.

No co-tenants to amortize a cluster's overhead against.

The framing fact

What LionX actually uses today.

LionX uses no Kubernetes feature.

operators

CRDs

service-mesh

StatefulSets

HorizontalPodAutoscaler

The current on-prem cluster is a delivery vehicle, not an architectural commitment.

Nothing in LionX pays for the Kubernetes API surface. Choosing Kubernetes on AWS would keep a cost whose benefit we never collect.

Three options

Same workload on each, judged on what costs engineer time.

	 ECS Fargate ✓	 ECS on EC2	 EKS standard to Auto Mode
KUBERNETES TO RUN	● none	● none	● cluster, upgrades, GitOps
NODE OPERATIONS	● AWS's job	● ours: AMIs, patching, CVEs	● ours, or AWS's with Auto Mode
AUDIT SURFACE	● CloudTrail only	● plus OS-level evidence	● plus Kubernetes audit logs
FIT FOR LIONX	● runs everything, nothing unused	● runs everything, ops attached	● pays for features nothing uses
FULLY LOADED, PER YEAR	● ~\$8K to 11K	● ~\$18K to 33K	● ~\$27K to 95K

The cheapest bill buys the most ops. Numbers on the next slide, full pros and cons in the ADR.

The numbers

Yearly, for the LionX footprint. Zurich list prices with a 1-year Savings Plan, ops deliberately underpriced at \$1,000 a day.

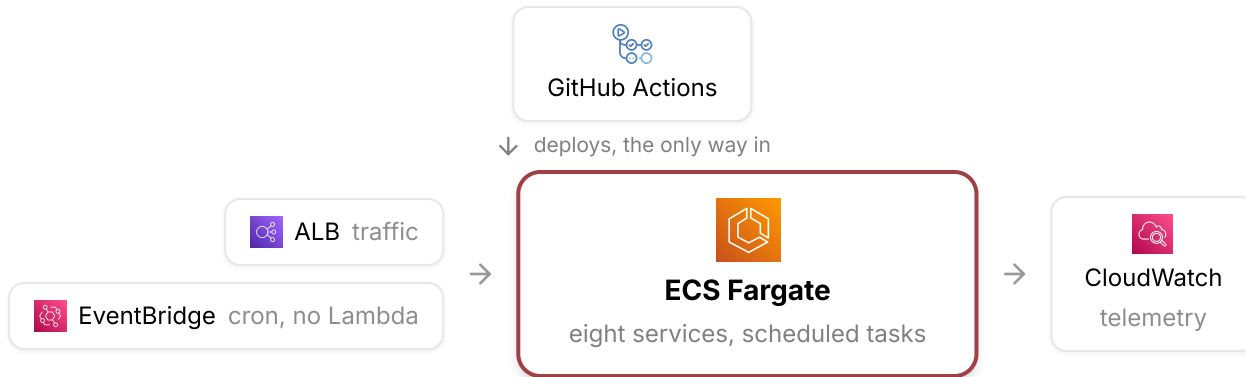
	 ECS Fargate ✓	 ECS on EC2	 EKS standard to Auto Mode
INFRA, PER YEAR	~\$4.6K	~\$3.1K	~\$4.3K to 5.2K
OPS, HOURS PER MONTH	~2 to 4	~10 to 20	~15 to 60
OPS, PER YEAR	~\$3K to 6K	~\$15K to 30K	~\$23K to 90K
FULLY LOADED, PER YEAR	● ~\$8K to 11K	● ~\$18K to 33K	● ~\$27K to 95K

Fully loaded, Fargate is the cheapest option, not the priciest. Ops hours decide it, the EKS range sits roughly 3 to 12 times above Fargate.

eu-central-2 list prices from the AWS Pricing API, cross-checked July 2026. Ops hours are assumptions from industry write-ups, not measurements, sourced and detailed per EKS variant in the ADR.

Decision: ECS Fargate

The smallest operational surface that runs all of LionX.



Zero Kubernetes to run

no cluster, no CNI, no GitOps reconciler,
no upgrade calendar

One audit surface

CloudTrail end to end, host patching is
AWS's responsibility

Nothing blocks landing

no cluster startup on the critical path to
end 2026

Accepted trade-offs. Each schedule is an EventBridge rule plus an IAM role, more Terraform than a CronJob. The ecosystem is narrower than Kubernetes, and nothing we run needs the difference.

Guardrails and next steps

How we keep the decision honest.

CONFIRMATION

- ✓ Every workload on Fargate, audited in the migration playbook. Nothing escapes to raw EC2 or EKS.
- ✓ CI is the only deployer. Manual service updates are flagged at PR review.
- ✓ CloudTrail retention matches Pictet audit policy.

FOLLOW-UPS

- 🚀 PoC in the new AWS account: pipeline, IAM boundaries and the cron path, end to end.
- 🔑 Revisit if phase 2 brings a hard need for mesh, operators or stateful Kubernetes.
- 👤 Revisit if a shared EKS platform lands at Pictet with co-tenants to split the cost.

Status: proposed. Comments on the PR, then we ratify.